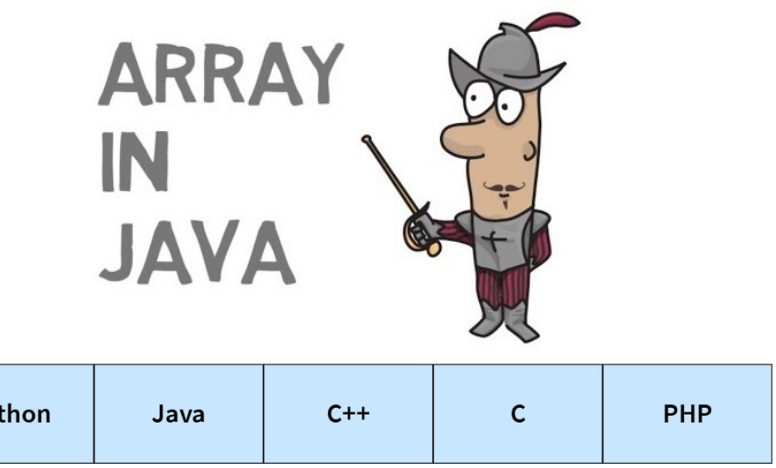
ARRAYS DATA STRUCTURE-BASICS



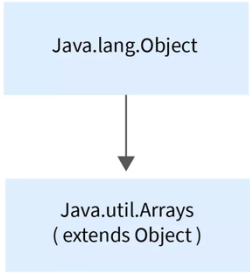
Python	Java	C++	C	PHP

Arrays- Introduction

Arrays are the most used **Data Structure.** Arrays in Java are **non-primitive** data types that store elements of a similar data type in the memory. Arrays in Java can store both primitive and non-primitive types of data in it.

Arrays inherit the Object Class and implements the serializable and cloneable interfaces.

The elements of an array are indexed, which means we can access them with numbers called **indices**. We can consider an array as a numbered list of cells, each cell being a variable holding a value. In Java, the numbering starts at 0.



Declaration and Creation of Arrays

Arrays Creation heap is a three step process:

1. Declaration of Array
2. Creation of Array
3. Initialisation/Accessing of Array

Declaration of Array

As discussed above, an array is a data structure or container that stores the **same typed** elements in a sequential format, so we need to declare the type first.

We can declare an array in three ways but we use mainly the first of the kind. Learn all of them at [https://replit.com/@ManuChaitanya/arraysDSBasics - Main.java](https://replit.com/@ManuChaitanya/arraysDSBasics#Main.java)

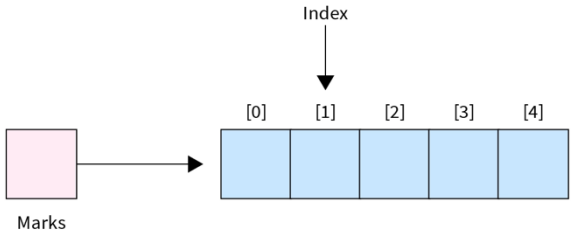
Creation of Array

In every program, to use an array, we need to create it before it is used. We require two things to create an array in Java - **Data type and Length.**

As we know, an array is a homogenous container that can only store the same type of data. Hence, it's required to specify the data type while creating it. Moreover, we also need to put the number of elements the array will store. As we discussed length of array are fixed, we fix it while creation, size data type can be integer, short, byte and char.

Unlike other programming languages, while declaring an array in java, we can put [] both before and after the variable name.

The **new keyword** is crucial to creating an array. Without using it, we cannot make array reference variables.

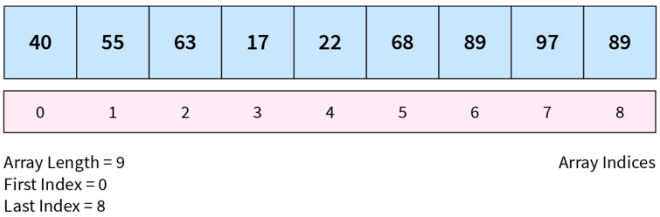


Initalization of Array

We can Initialize Arrays in Java in two Different ways.

1. Initialize using index - Once an array is created successfully, we can assign the value to any specific element using its index. Un-assigned elements automatically take the default value of the array's data type. In this case, 0 for the integer data type. On the other hand, if we try to enter more values than the size of the array, it will raise an ArrayIndexOutOfBoundsException.

2. Initialize while declaring - We can also initialize the whole array while declaring it just by using curly brackets {}. The string must be in quotes, whereas int, float, etc., do not require that.

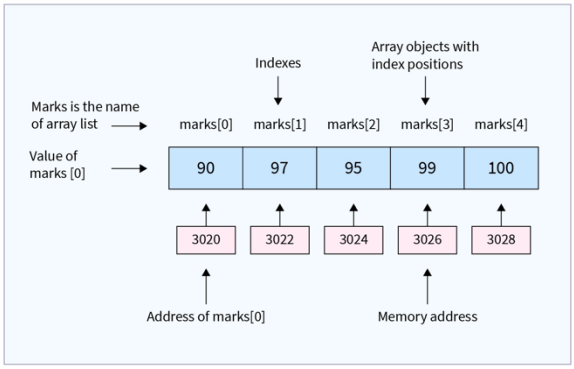


How to Change an Array Element Accessing it?

As we have already discussed Arrays Objects are **Mutable** thus we can change the internal structure of the object unlike Strings. To change a value at a specific position in Java arrays, refer to an index. By providing index value, we tell the Java compiler where we want to change the value.

For this, we need to use a Square bracket [] with an index number inside it. If somehow we entered the index number bigger than the array length, then an exception occurs of type ArrayIndexOutOfBoundsException.

Memory address for array marks -



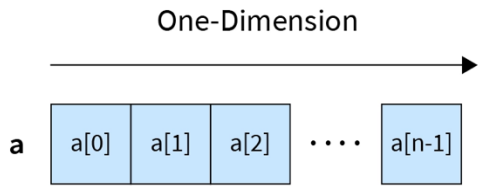
TYPES OF ARRAYS

In Java, there are two types of Arrays:

1. Single-Dimensional Array
2. Multi-Dimensional Array

Single Dimensional Array

An array that has only one subscript or one dimension is known as a single-dimensional array. It is just a list of the same data type variables. One dimensional array can be of either one row and multiple columns or multiple rows and one column.

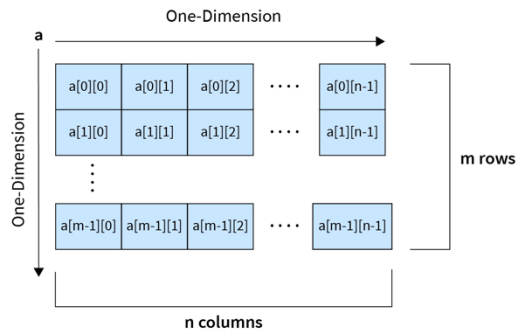


Multi-Dimensional Array

In not all cases, we implement a single-dimensional array. Sometimes, we are required to create an array within an array.

A multi-dimensional array is just an array of arrays that represents multiple rows and columns. These arrays include **2D, 3D, and nD types**. 2D data like tables and matrices can be represented using this type of array.

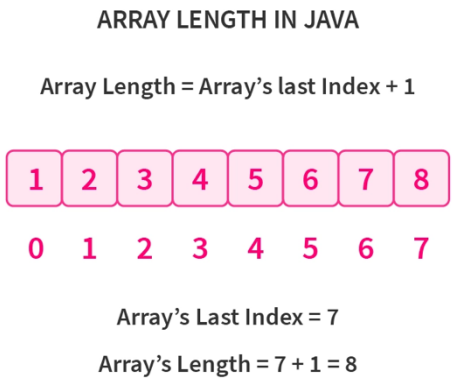
To implement a multi-dimensional array, one needs to add two square brackets instead of one.



If we want to change any specific value, we need to use two square brackets, the first representing the rows and the second representing the columns. Check Creation of Multi-Dimensional Arrays in notes.

https://replit.com/@ManuChaitanya/arraysDSBasics#Main.java

Length of Arrays

In **Java,** the array length is the number of elements that an array can holds. There is no predefined method to obtain the **length of an array**. We can find the **array length in Java** by using the array attribute **length**. We use this attribute with the array name.

Every array has an in-built **length** property whose value is the size of the array. Size implies the total number of elements that an array can contain. The length property can be invoked by using the dot (.) operator followed by the array name.